

CSCE 52803: Assignment 3

Due 11:59 pm Saturday, October 10, 2026

1. Consider the **Unbounded** Knapsack Problem: given a knapsack of capacity S , n items of volumes v_i and weights w_i , find the packing with the maximum total weight.

(1) Design a greedy algorithm for solving the Knapsack Problem that considers the weight per volume.

(2) Provide an example showing that the solution produced by the greedy algorithm may not be the optimal solution.

(3) Suppose that $\forall i, v_i \leq S$. Show that the approximation ratio of the greedy algorithm is $\frac{1}{2}$.

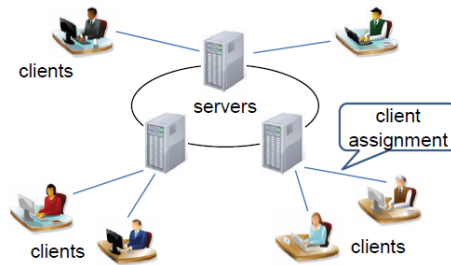
2. Consider the Vertex Cover Problem: given an undirected graph $G = (V, E)$ where $V = \{v_i\}$ is a set of vertices and $E = \{e_j\}$ is a set of edges, find the minimum vertex cover.

(1) What is the integer linear programming problem for formulating the Vertex Cover Problem? What is the relaxed linear programming problem?

(2) Denote the optimal solution of the relaxed linear programming problem as $\{x_i^*\}$. What is the procedure of randomized rounding based on $\{x_i^*\}$ to obtain the solution C to the Vertex Cover Problem?

(3) Let OPT be the size of the optimal solution. Show that $E(|C|) \leq OPT$ and for each $e_j \in E$, $\Pr(C \text{ covers } e_j) \geq \frac{3}{4}$.

(4) Based on solution $\{x_i^*\}$, how to obtain a solution C' to the Vertex Cover Problem such that $E(|C'|) \leq \log_2 m \cdot OPT$ and $\Pr(C' \text{ is a vertex cover}) \geq 1 - \frac{1}{m}$, where m is the number of edges?



3. Consider the Client Assignment Problem described in Assignment 2: given a weighted graph $G = (V, E)$ where V contains a set of clients C and a set of servers S , and a weight/latency $d(u, v)$ is associated for each edge $(u, v) \in E$, find a client assignment $s: C \mapsto S$ that minimizes the maximum latency between any pair of clients, i.e.,

$$\text{minimize } \max_{c_i, c_j \in C} L(c_i, c_j)$$

where

$$L(c_i, c_j) = d(c_i, s(c_i)) + d(s(c_i), s(c_j)) + d(s(c_j), c_j)$$

Suppose that the latency satisfies the triangle inequality, i.e., for any vertices $a, b, c \in V$, we have $d(a, b) \leq d(a, c) + d(b, c)$.

(1) Consider a simple algorithm that assigns each client to its nearest server. Show that the approximation ratio of this algorithm is 3.

(2) Given a tight example of the above approximation ratio.

(3) Design a greedy algorithm that assigns clients to servers iteratively. Observe the fact that whenever a client c is assigned to a server s , then all unassigned clients that are not further from s than c can also be assigned to s without increasing the objective, i.e., $\max_{c_i, c_j \in C} L(c_i, c_j)$. Based on this observation, design a greedy algorithm where in each iteration it selects one client to be assigned to a server such that the increase in the objective per client is minimal.